PAPER_236

PAPER<i>236: UQFF Learning Assessment Evolution</i>B — Framework Meta-Assessment Advancement Score

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Session: *59 (grokshare8d951e12.txt second-pass — Doc 9)*

PAPER*236: UQFF Learning Assessment Evolution*B — Framework Meta-Assessment Advancement Score

Author: Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grok*share*8d951e12.txt second-pass — Doc 9) **Date:** March 2026 **Classification:** Novel — First Framework-Level Meta-Assessment Calculator in UQFF Pipeline **Status:** Proof-Quality Whitepaper **CP3 Class:** `UQFFLearningAdvancementCalculator`

Abstract

This paper documents the UQFF Learning Assessment Evolution_B module, the ninth in a series of 500+ UQFF code files. Unlike all preceding modules (which target specific astrophysical objects), this is the first **framework-level meta-assessment calculator** in the CP1/CP2/CP3 pipeline. It computes a structured advancement score aggregating three orthogonal metrics: physical regime diversity, dynamical term novelty, and framework scalability. The advancement formula is:

$$\text{advancement} = \frac{\text{diversity\_score} + \text{dynamic\_score} + \text{scalability\_score}}{3.0} \times 100\%$$

The module aggregates parameter inputs from three consecutive UQFF examples: Westerlund 2 stellar wind, Pillars of Creation erosion, and Rings of Relativity lensing modulation — providing a comparative multi-system basis for evaluating framework maturity.

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1. Module Context

Source: `UQFFLearningAssessment.h` (Evolution_B, October 08 2025 revision) Integration target: `ziqn233h.cpp` (base program, not included) Designed to instantiate as: `UQFFLearningAssessment assess; double adv = assess.compute_advancement();`

This is the **first CP3 calculator not modelling an astrophysical object** — it models the UQFF framework's own learning progression, providing a quantitative measure of advancement per development session.

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2. Core Advancement Formula

$$\boxed{\text{advancement} = \frac{d + D + s}{3.0} \times 100\%}$$

where:

- `d` = `diversity_score` — number of distinct physical regimes demonstrated (default 3: stellar wind, erosion, lensing)

- D = dynamic_score — number of novel dynamic force/field terms introduced (default 3: $a_{\rm wind}$, $E(t)$ erosion, lensing modulation)
- S = scalability_score — adaptability across spatial/temporal scales, normalised to [0, 1] (default 0.8)

Default result:

$$\text{advancement} = \frac{3.0 + 3.0 + 0.8}{3.0} \times 100\% \approx 226.7\%$$

(Values > 100% indicate multi-regime simultaneous coverage, i.e., framework is demonstrating super-linear progression.)

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3. Parameters from Three Prior Examples

3.1 Westerlund 2 (Stellar Wind Regime)

Parameter	Symbol	Default Value	Units
Initial mass	$M_{\rm wd2}$	$30{,}000\,M_\odot = 5.967\times10^{34}$	kg
Radius	$r_{\rm wd2}$	(set per session)	m
Star formation timescale	$\tau_{\rm SF,\,wd2}$	3.15×10^{13}	s
Wind density	$\rho_{\rm wind,\,wd2}$	1×10^{-20}	kg/m^3
Wind velocity	$v_{\rm wind,\,wd2}$	$2{,}000\times10^3$	m/s

3.2 Pillars of Creation (Erosion Regime)

Parameter	Symbol	Default Value	Units
Erosion factor (initial)	E_0	0.3	—
Erosion timescale	$\tau_{\rm erosion}$	3.15×10^{12}	s (~0.1 Myr)

Erosion temporal model: $E(t) = E_0\,e^{-t/\tau_{\rm erosion}}$

3.3 Rings of Relativity (Gravitational Lensing Regime)

Parameter	Symbol	Default Value	Units
Einstein radius	$r_{\rm rings}$	1.54×10^{22}	m
Lensing amplitude factor	$L_{\rm factor}$	1.2	—
Hubble parameter at z	H_z	2.18×10^{-18}	s^{-1}

Lensing modulation: $g_{\rm lens} = U_{g1}\cdot H_z\cdot t + U_{g4}(1+f_{\rm TRZ})\cdot L_{\rm factor}$

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4. Setter Architecture

The original C++ header provides per-variable setters:

```
void setDiversityScore(double v) / addToVar / subtractFromVar — same pattern for all params
```

This enables runtime update of any parameter without reconstructing the object, supporting iterative UQFF refinement workflows.

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5. Novel Contribution

- **First meta-assessment calculator** — evaluates UQFF progression itself, not an astrophysical object
- **Three-metric advancement formula** — diversity × dynamic × scalability normalised mean
- **Multi-example parameter aggregation** — single class absorbs parameters from 3 prior sessions
- **Advancement values > 100%** — framework intentionally designed to go beyond binary pass/fail
- **Supports quantitative session comparison** — advancement score can be tracked per git commit

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6. CP3 Implementation

Class: `UQFFLearningAdvancementCalculator` **Method:** `compute(dataset) → dict` **Key output:** `advancement_pct` (float, %) **Available equations:** wind acceleration, erosion decay, lensing modulation, advancement formula

```
calc = UQFFLearningAdvancementCalculator()
result = calc.compute({'diversity_score': 3.0, 'dynamic_score': 3.0, 'scalability_score': 0.8})
# result['advancement_pct'] ≈ 226.7 %
```

References

- Murphy, D.T. (2025). *UQFF Learning Assessment Evolution_B Module*, `UQFFLearningAssessment.h`, October 08 2025
- grok*share*8d951e12.txt, Doc 9, lines 2993–3085 (UQFF Learning Assessment Evolution_B)
- Westerlund 2: PAPER_228 (`Westerlund2MUGEStellarWindCalculator`)
- Pillars of Creation: PAPER_229 (`PillarsOfCreationErosionMUGECalculator`)
- Rings of Relativity: Prior session `UQFFLensingModulationRingsCalculator`
